

Physics-Guided Generative Design for CDK2 Selectivity

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Potency ≠ Selectivity

Related kinases share similar ATP-binding sites, so a molecule predicted to bind CDK2 may also bind unwanted kinases.

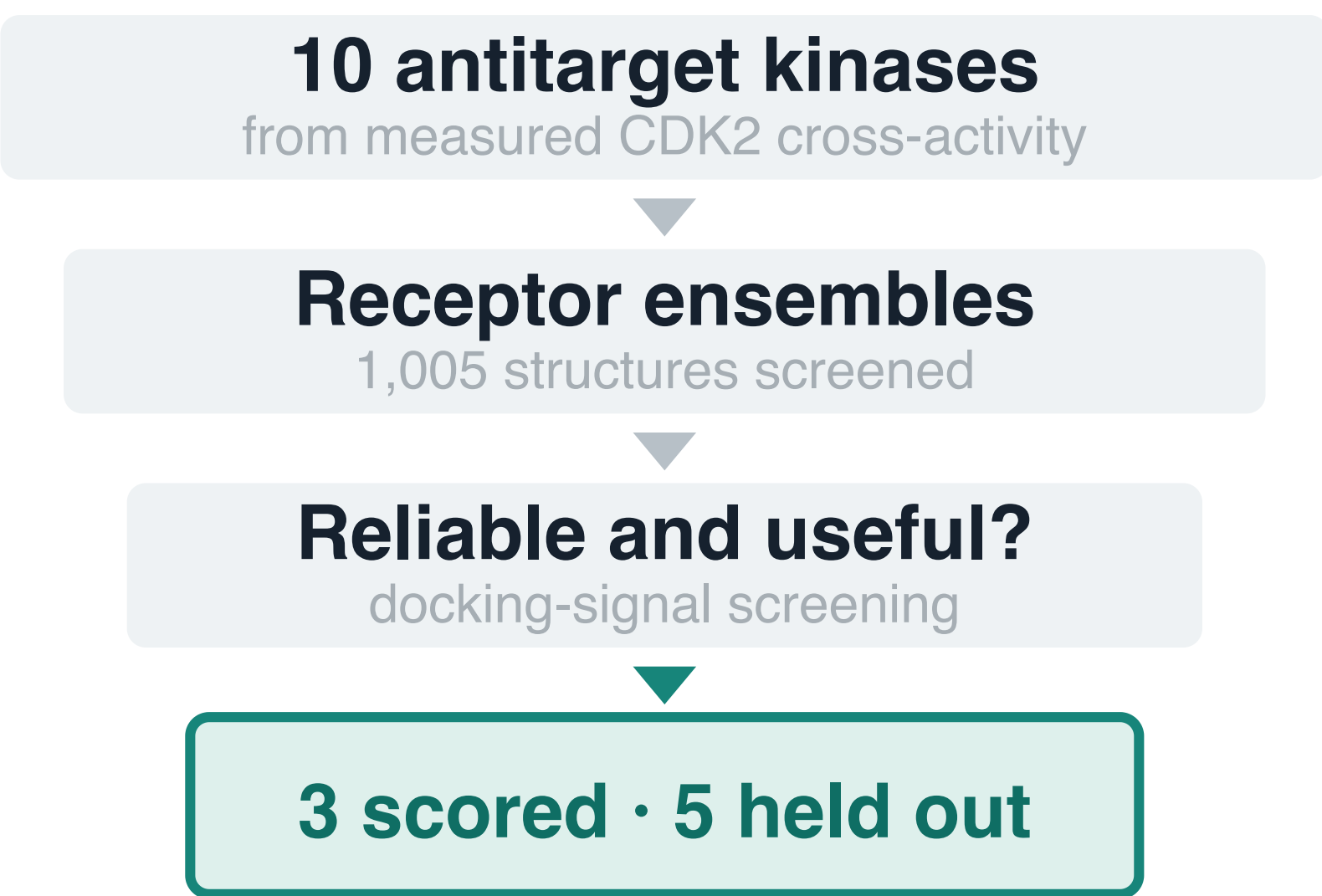


Can ALGen preserve predicted CDK2 binding while avoiding related kinases?

Tested here through docking-guided molecular selection

METHOD

Which off-targets can we trust?



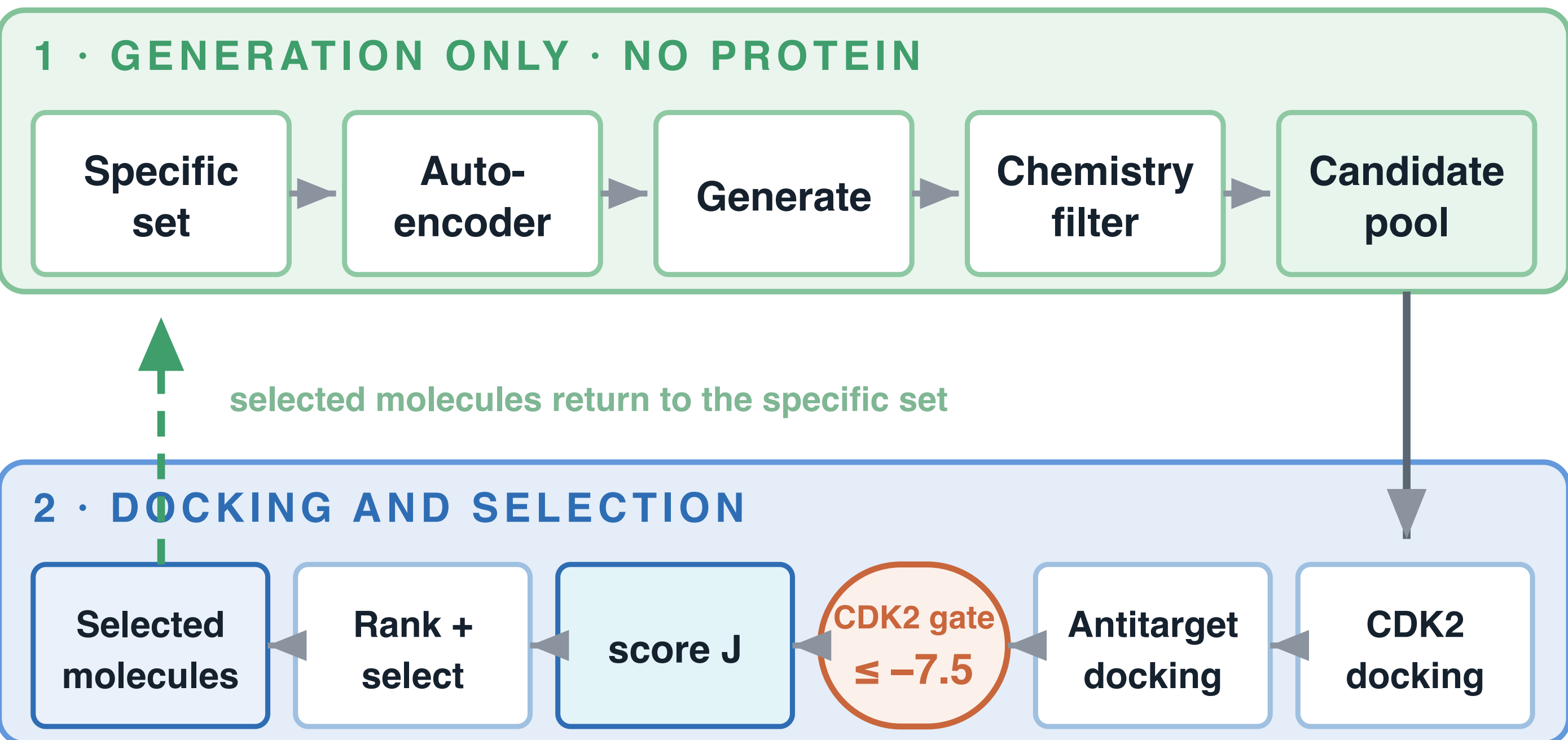
RELIABLE → separates known binders

USEFUL → adds selectivity information

SCORED	DYRK1A · GSK3B · PIM1
HELD OUT	CDK1 · CDK5 · CDK6 · CDK7 · CDK9

scored enter J · held-out only evaluated

How does ALGen use them?



Docking guides selection, not gradients.

Protein information enters only through the molecules selected and recycled.

What does J reward?

- + TARGET
strong CDK2 binding
- ANTITARGET
binding to unwanted kinases
- PROMISCUITY
broad non-specific chemistry
- + SIZE CONTROL
prevents trivial shrinking

$$J = \text{target} - \beta_{\text{AT}} \cdot \text{antitarget} - \text{promiscuity} + \text{size}$$

$\beta_{\text{AT}} = 1$ ON $\beta_{\text{AT}} = 0$ OFF

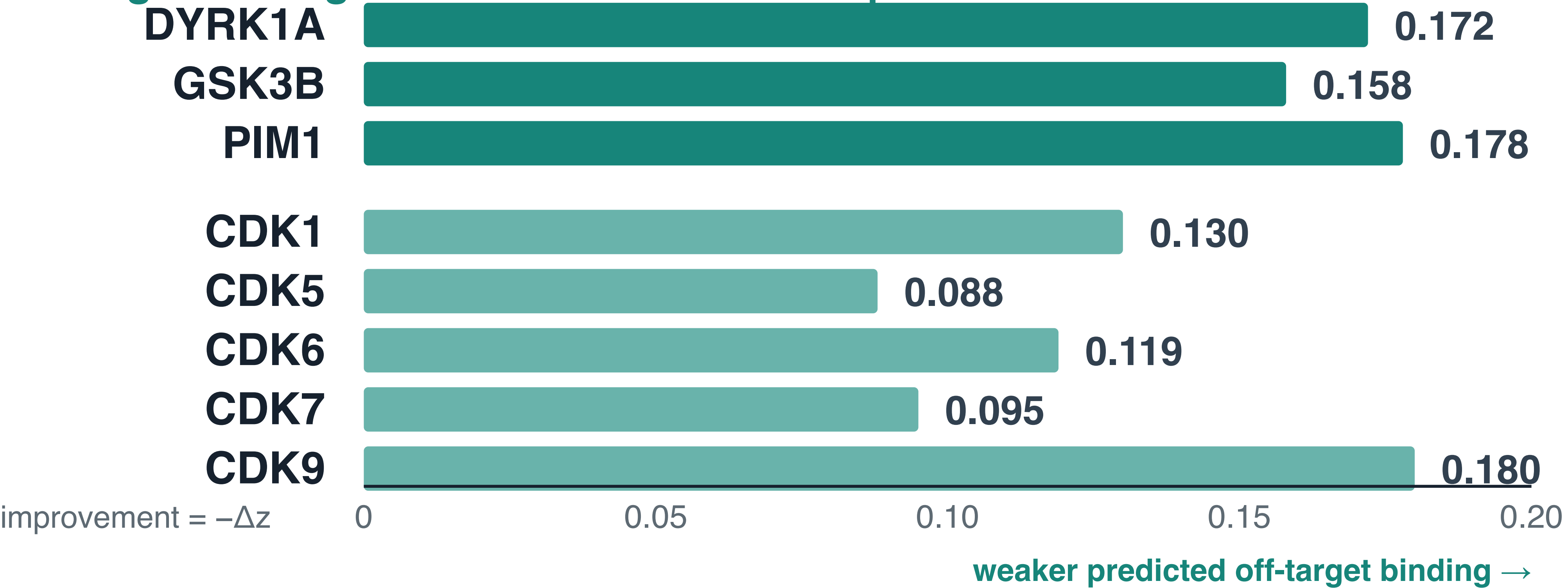
RESULT

SELECTION-LEVEL PROOF OF CONCEPT ✓

Antitarget guidance changes molecular selection

Same candidate pool. Same selection budget. Only β_{AT} changes.

Off-target binding weakens across all 8 receptors



8 / 8

RECEPTORS IMPROVE
all 8 reported intervals exclude zero

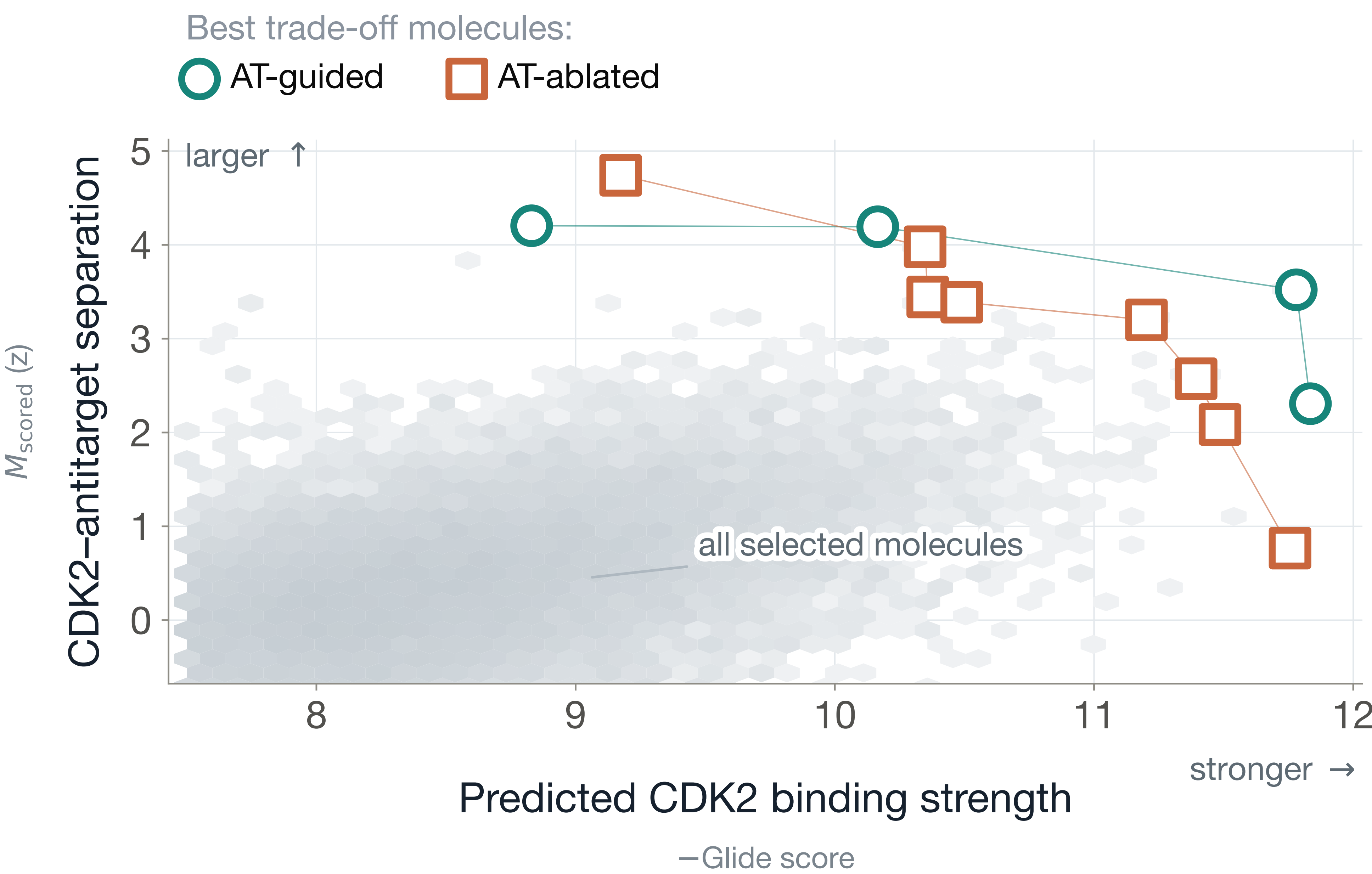
13.95%

SELECTION CHANGED
279 of 2,000 from one 26,062-molecule pool

-0.058 z

CDK2 COST
slightly weaker predicted target binding

Better potency–selectivity trade-offs



Take-home

✓ Selection shifts

AT guidance changes which molecules survive.

✓ 8 / 8 off-targets improve

with a small predicted CDK2 cost.

→ Next: replicated trajectories

test persistence through later generation.

-0.099 z observed ≈ ±0.102 z trajectory variability · 1 trajectory per condition

